

num  
10

if (num == 10)

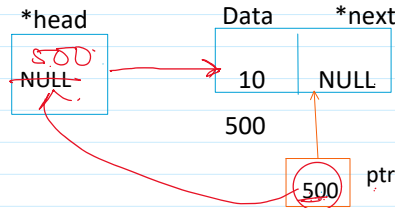
① Create a node.

② Assign the value to the data field.  
 $ptr \rightarrow data = value;$

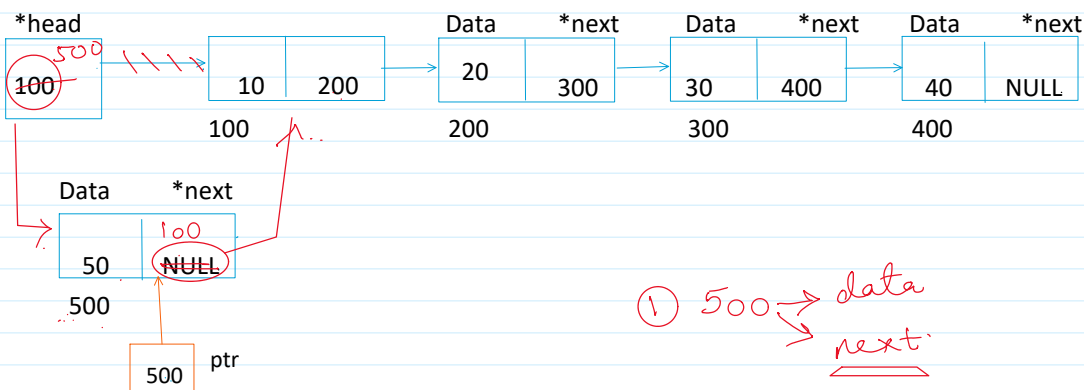
③ Attach the node to the linked list.

a) If the linked list is empty.

if (head == NULL)  
head = ptr;



b) If linked list contains nodes



①  $500 \rightarrow data$   
 $500 \rightarrow next$

$500 \rightarrow next = 100;$   
 $ptr \rightarrow next = head;$

- 1) Connect the new node to the first node of the list.
  - a.  $Ptr \rightarrow next = head;$
- 2) Update the head pointer to point to the new node.
  - a.  $Head = ptr;$

②  $head = 500;$   
 $head = ptr;$

Best Case : If the list is empty :  $O(1)$

Worst / avg : if the list has multiple node :  $O(1)$